

DRAWING MACHINE

Creating the String artist

The pseudocode on the previous page plans a function called the String artist, which will turn a string of values into single commands that are sent to the Turtle controller. The next stage is to turn the pseudocode into real Python code, using a function called “split()”.

6 The “split()” function splits a string into a list of smaller strings. Each break point is marked by a special character (“-” in this program).

This string lists the commands to create the sample house shape

```
>>> program = 'N-L90-F100-R45-F70-R90-F70-R45-F100-R90-F100'
>>> cmd_list = program.split('-')
>>> cmd_list
['N', 'L90', 'F100', 'R45', 'F70', 'R90', 'F70', 'R45', 'F100', 'R90', 'F100']
```

The “split()” function breaks the string down into a list of separate commands

7 Now write out the pseudocode for the String artist using real Python code. Use the “split()” function to slice up the input string into turtle commands.

Tells the program to split the string wherever it sees a “-” character

```
def string_artist(program):
    cmd_list = program.split('-')
    for command in cmd_list:
        cmd_len = len(command)
        if cmd_len == 0:
            continue
        cmd_type = command[0]
        num = 0
        if cmd_len > 1:
            num_string = command[1:]
            num = int(num_string)
        print(command, ': ', cmd_type, num)
        turtle_controller(cmd_type, num)
```

This makes the program loop through the list of strings—each item is one command for the turtle

If the length of the command is 0 (so the command is blank), the function skips it and moves to the next one

Takes the first character of the command (remember, strings start at 0) and sets it as the command type (“F”, “U”, etc.)

This takes all the remaining characters from the command by cutting off the first one

Prints the command on the screen so you can see what the code is doing

Passes the command to the turtle

Gets the length of the command string

Converts the characters from strings into numbers

Checks if the command is followed by more characters (the number)